Actuator-Aware Geometric Flight Control of a Quadrotor

MCAV project — Technical report — October 2026

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This project delivers a flight controller for a 0.8 kg quadrotor that takes off and tracks a circular trajectory in the MuJoCo physics engine. The controller works directly on rotation matrices (no Euler-angle singularities). Its gains are not hand-tuned: they are computed by convex optimisation (linear matrix inequalities) so that every motor stays inside its 0–8 N range. On the nominal mission the tracking error converges to 2×10^{-9} m and the motor forces stay between 1.72 and 2.65 N. With gains taken from the literature, the same mission would need between -36 and 46 N per motor. A Monte Carlo campaign over 500 random initial positions and attitudes (up to 175°) converges in 95.6 % of the cases. Every failure there comes from hitting the ground, which the analysed model does not contain. Stability claims are checked by a computer-algebra system instead of being asserted by hand, and every number in this report is regenerated by one command.

Key result	Value
Steady-state tracking error	2×10^{-9} m (settles below 5 cm in 8.7 s)
Motor forces, nominal mission	1.72–2.65 N, inside the 0–8 N motor range
Same mission, gains from the literature	-36 to 46 N, not realisable
Monte Carlo (500 runs, attitude errors up to 175°)	95.6 % converge; 97.8 % never leave the motor range
Inside the certified attitude region (138 runs)	100 % converge, as the analysis predicts
Symbolic identities verified by SymPy	21
Gain synthesis (6 per-axis LMI designs, open-source solver)	0.7 s
Full reproduction of every result	one command, 149 s on a laptop

1 Problem and requirements

A quadrotor of mass 0.8 kg must take off from rest at $(0, 0, 0.5)$ m and track the horizontal circle

$$p_d(t) = [\,3\cos(0.5\,t),\ 3\sin(0.5\,t),\ 3\,]^\top \text{ m},$$

heading along the tangent. Each of its four motors delivers a thrust between 0 and 8 N (hover needs 1.96 N per motor). The physics is integrated by MuJoCo [7] at 500 Hz, and the controller runs at the same rate.

ID	Requirement	How it is met
R1	Zero steady-state tracking error	Exact feedforward of the reference acceleration and of the desired-frame motion (Sec. 3)
R2	Motor forces inside 0–8 N on the nominal mission	Gains synthesised from the motor budget by LMIs (Sec. 4)
R3	Every stability claim backed by checkable evidence	Proofs verified by SymPy, certificates re-checked by tests (Sec. 5)
R4	Robustness to the initial condition quantified	Monte Carlo over position and attitude errors (Sec. 6)
R5	Reproducible by a third party with one command	`uv run python main.py` (Sec. 9)

2 Architecture

The controller is a two-level cascade (Fig. 1). The outer loop turns the position error into a desired thrust vector. Its direction, together with the desired heading, defines the desired attitude R_d. The inner loop computes the torques that rotate the body onto R_d. A fixed allocation matrix then converts total thrust and torques into the four motor forces.

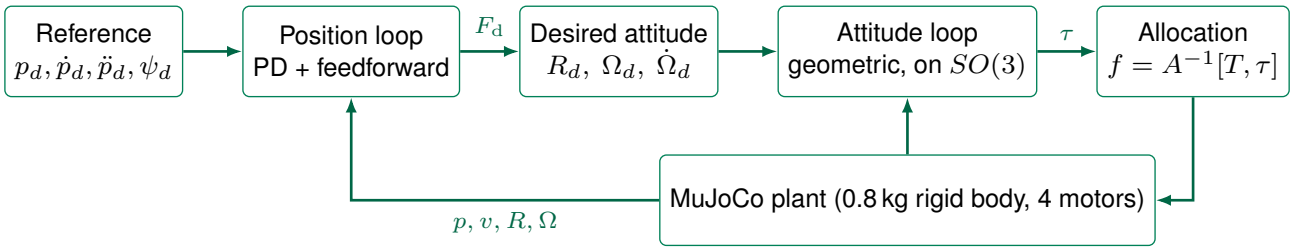


Figure 1: Control architecture. F_d: desired thrust vector; its magnitude T goes directly to the allocation.

Software. The guiding rule was to write only the parts that are specific to this problem and to rely on maintained libraries for everything else (Table 1). Vehicle parameters have a single source of truth: mass, inertia, motor positions, rotor torque coefficient and motor range are read from the MuJoCo model file. The allocation matrix in particular is MuJoCo's own actuator transmission, not a hand-written matrix.

Need	Library	Role
Rigid-body physics, allocation	MuJoCo	Simulation; mass, inertia and allocation read from the model
Symbolic verification	SymPy	Derives and checks every identity of the stability analysis
Gain synthesis	cvxpy + Clarabel [3]	Semidefinite programs, open-source solver, no licence
Rotations, ODE integration	SciPy	Attitude sampling, unicycle integration (RK45)
Parallel Monte Carlo	joblib	One rollout per core
Data, plots, tests	pandas, matplotlib, pytest	Logs, figures, 29 automated tests
Environment	uv	Locked, reproducible dependency set

Table 1: Off-the-shelf components. The project-specific code (controller, gain design, analysis, experiments) is about 1 100 lines of Python.

3 Control design

The design follows the geometric tracking controller of Lee et al. [6], with the weighted attitude error of [5] so that every axis can get its own gain.

Position loop. With $e_p = p - p_d$ and $e_v = \dot{p} - \dot{p}_d$,

$$F_\mathrm{des} = m\big(-K_v K e_p - K_v e_v + \ddot{p}_d + g e_3\big), \qquad T = \|F_\mathrm{des}\|. \tag{1}$$

The feedforward term $\ddot{p}_d$ supplies the centripetal acceleration. Without it, the vehicle settles on a smaller, lagging circle, with a 17 cm error in the previous version of this project.

Attitude loop. R_d has its third axis along F_des and its first axis as close as possible to the heading. With the attitude error $R_e = R_d^\top R$,

$$\tau = \underbrace{-e_R - K_\Omega e_\Omega}_{\text{feedback}} + \underbrace{\Omega \times J\Omega - J\big(\hat{\Omega} R_e^\top \Omega_d - R_e^\top \dot{\Omega}_d\big)}_{\text{cancellation}}, \qquad e_R = \tfrac{1}{2}\big(G R_e - R_e^\top G\big)^\vee. \tag{2}$$

The cancellation terms remove the gyroscopic coupling and the motion of the desired frame. Without them the residual error does not vanish (1.3 mm RMS on the nominal mission) and the stability proof does not apply. The desired angular velocity Ω_d and its derivative come from the sequence of desired attitudes by finite differences. On the steady circle this is exact; during transients it is accurate to first order in the 2 ms control period.

4 Actuator-aware gain synthesis

Gains published for this controller were designed for a 4.3 kg vehicle with an inertia about 20 times larger. On this vehicle they demand -36 to 46 N per motor (Fig. 2, left), far outside what the motors can deliver. Rather than re-tuning by trial and error, the gains are computed from the motor budget.

Budget. Each motor force is $f_i = T/4 + \delta_i$, where δ_i is set by the torques. Reserving 1.5 N of each motor's range for torques, and splitting it among roll, pitch and yaw, gives torque budgets of 0.234 N m (roll, pitch) and 0.0288 N m (yaw: the rotor drag torque is small, so yaw authority is about ten times weaker). The remaining range bounds the correction acceleration to 3.0 m/s^2 horizontally and 2.31 m/s^2 vertically. If every loop respects its budget, every motor stays inside 0–8 N.

Synthesis. For each axis of each loop (a double integrator once the cancellation terms act) we solve the input-bounded state-feedback problem [2]: find $W \succ 0$ and X with $K = XW^{-1}$ such that

$$AW + WA^\top + BX + X^\top B^\top \preceq -2\alpha W, \qquad \begin{bmatrix} 1 & x_0^\top \\ x_0 & W \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} \bar{u}^2 & X \\ X^\top & W \end{bmatrix} \succeq 0, \tag{3}$$

and maximise the decay rate α by bisection. Every trajectory of the linear model that starts in the ellipsoid $\{x^\top W^{-1} x \leq 1\}$ stays in it, decays at rate α, and never asks for more than $\bar{u}$. The design points x_0 are the mission start plus a margin. The result is a position loop with rates of 0.53–0.78 s^{-1} and an attitude loop at 9.1 s^{-1} (roll, pitch) and 2.1 s^{-1} (yaw). The inner loop is therefore faster than the outer loop, as the cascade needs. The nominal mission then stays between 1.72 and 2.65 N per motor (Fig. 2, right), at the cost of a slower transient: 8.7 s instead of 3.3 s to reach 5 cm.

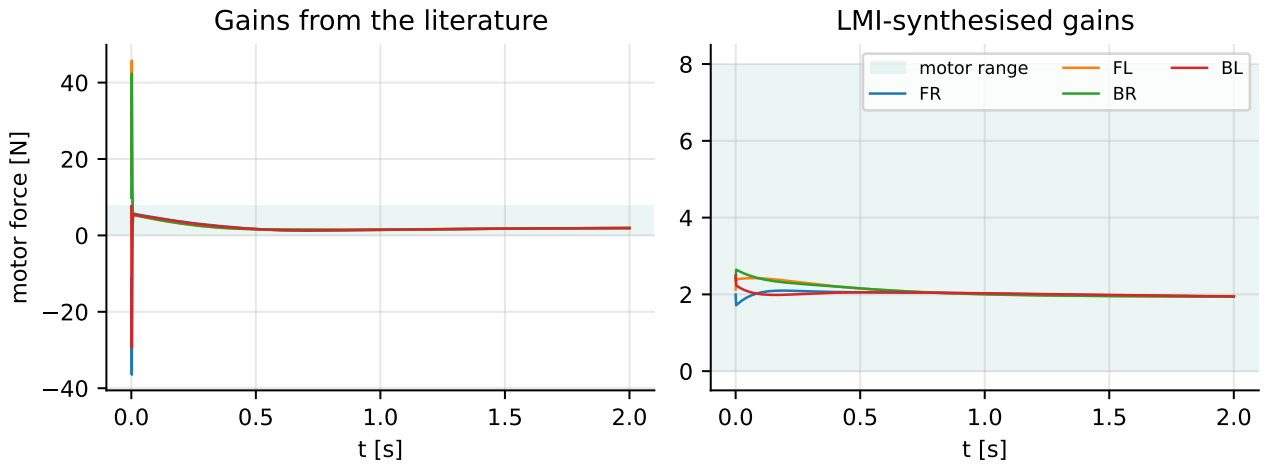


Figure 2: Motor forces during take-off. Left: gains from the literature. Right: LMI gains. Shaded: motor range.

5 Verification and validation

The project separates what is proved, what is checked numerically, what is observed in simulation and what is assumed (Table 2).

What is not claimed. The closed loop is not globally asymptotically stable. No continuous feedback on the rotation group can be [1]: a 180° attitude error is an equilibrium of any such controller. The analysis states its region explicitly instead.

Symbolic verification. The stability argument rests on a few exact identities: the attitude error function decreases at the rate $\dot{\Psi} = e_R \cdot e_\Omega$; the cancellation terms make the attitude error dynamics independent of the trajectory; and the only attitudes with $e_R = 0$ are the identity and 180° rotations. SymPy derives each identity from the equations of motion and the control law, and checks it for a generic matrix, so no small-angle approximation is involved. It also produces a concrete state at which the feedback-only law makes the Lyapunov function increase. That state shows why the cancellation terms are needed.

Automated tests. 29 `pytest` tests cover the symbolic identities, the LMI certificates (decay rate, design points inside the ellipsoid, input bound over the ellipsoid), and the closed loop. Among them: the mission converges, all

Status	Statement	Evidence
Proved	The attitude error converges from every state with $V_2 < 0.092$, i.e. roll/pitch errors up to about 37° and any yaw error below 180° at zero initial rate	Lyapunov function and LaSalle; 21 identities verified by SymPy
Proved	The full closed loop is locally exponentially stable	Cascade structure and linearisation [4]
Proved	On the reduced linear models, the LMI gains respect the motor budget inside the certified ellipsoids	LMI certificates, re-checked by tests independently of the solver
Observed	Convergence and motor feasibility from large initial errors	MuJoCo simulation, Monte Carlo
Assumed	Motors can produce any force (no saturation in the analysis); $\Omega_d, \dot{\Omega}_d$ exact (implemented by finite differences)	Feasibility is reported when the range would be exceeded

Table 2: Evidence behind each claim.

motor forces stay in range, the Lyapunov function never increases along the simulated trajectory, the finite-difference Ω_d matches its exact steady-state value, and the feedback-only law leaves a residual error.

6 Results

Nominal mission. The vehicle converges to the circle and the error decreases to machine precision (Fig. 3). The feedback-only law, without the cancellation terms, stalls at the millimetre level. The attitude Lyapunov function never increases along the simulated trajectory (Fig. 5), as the analysis predicts.

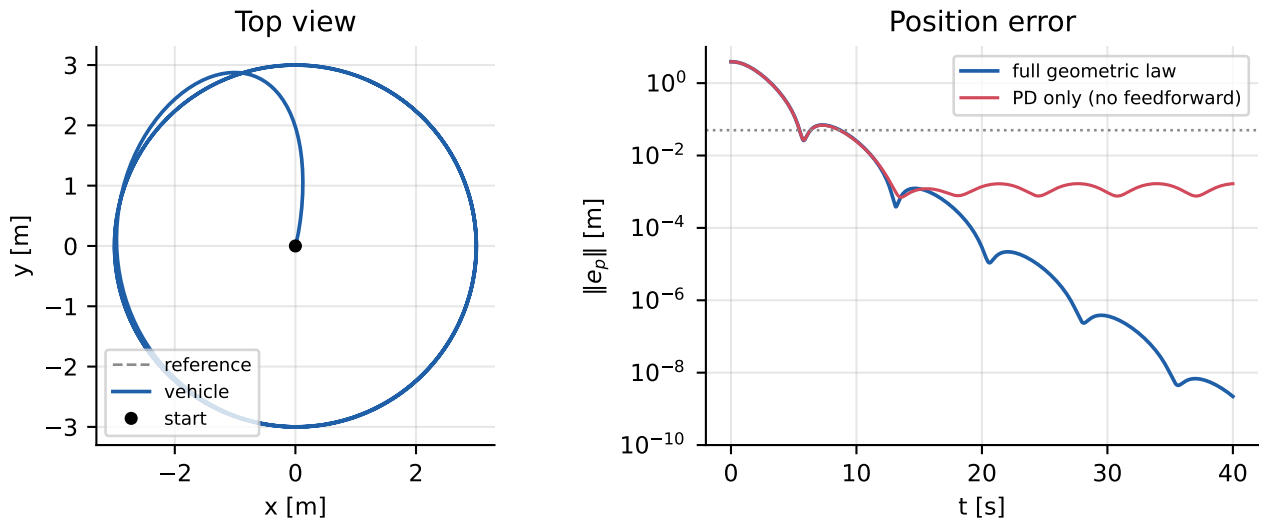


Figure 3: Nominal mission with the LMI gains. Left: top view. Right: position error, with and without the cancellation terms; dotted line at 5 cm.

Robustness to the initial condition. 500 runs start from random position offsets (up to 2 m) and random attitude errors (up to 175° about a random axis), at rest, and fly the same mission with the same gains (Fig. 4). 95.6 % converge. The other 4.4 % hit the ground during the recovery manoeuvre; the floor exists in the simulation but not in the analysed model. All 138 runs that start inside the certified attitude region converge, as the analysis predicts. 97.8 % of all runs never need more than the motors can deliver, even though the LMI certificate covered only the nominal take-off envelope.

7 Engineering review: defects found and fixed

A systematic review of an earlier version of the project found the defects below. Each one is now covered by a fix and, where possible, by an automated test.

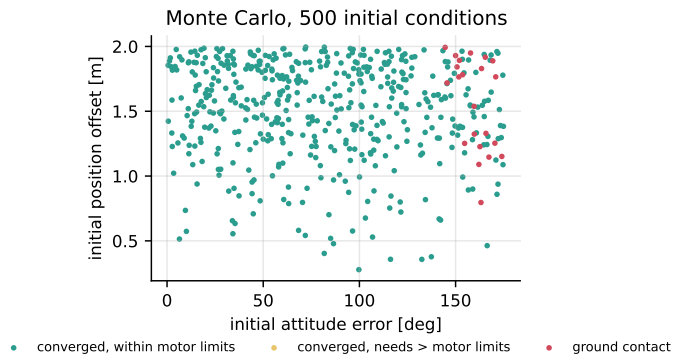


Figure 4: Monte Carlo: outcome for each initial condition.

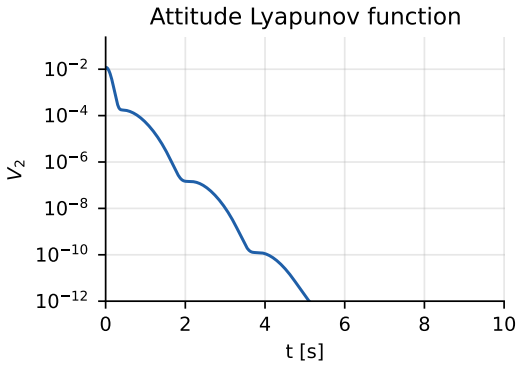


Figure 5: Attitude Lyapunov function on the nominal mission.

Defect	Measured impact	Fix
Reference acceleration not fed forward	17 cm steady-state error	Exact $\ddot{p}_d$ from the reference
Moment arm set to the arm length L instead of $L/\sqrt{2}$	Roll/pitch torques realised at 71 %	Allocation read from the MuJoCo model
Simulator silently clipped motor commands, while the analysis assumed no limits	Simulated and analysed systems differed	Limits stored in the model, enforced as a design constraint, reported in validation
Desired angular velocity approximated by the yaw rate	3 cm residual error even with the acceleration fixed	Ω_d, $\dot{\Omega}_d$ from R_d
Gains copied from a 4.3 kg vehicle	Peaks above 150 N per motor	LMI synthesis from the motor budget
"Globally asymptotically stable" claimed with a hand-written derivative	Unsupported, and impossible on $SO(3)$	Verified local results with explicit regions
About 900 lines of custom simulation, logging and plotting code	Maintenance cost	Replaced by MuJoCo, SciPy, joblib, pandas

8 Limitations and next steps

- **Actuator saturation.** Motor limits are respected by design on the nominal mission, but are not enforced online. Next step: a control barrier function or a model predictive controller acting on the same allocation.
- **Unmodelled effects.** Motor dynamics, delays, aerodynamic drag, ground effect and sensor noise are not simulated. The state is assumed to be measured exactly. Next step: an estimator (EKF on IMU data) and a robustness study.
- **Path to hardware.** The controller is a pure function of state and reference at a fixed rate. It can be ported to a PX4 or ROS 2 software-in-the-loop setup before any flight test.

9 Reproducing the results

The only prerequisite is `uv`, which installs Python and the locked dependency set.

`cd mcav_project && uv run python main.py` — runs the symbolic checks, the LMI design, the simulations and the Monte Carlo, and writes the figures and `numbers.tex` used by this report (149 s).

`uv run pytest` — runs the automated tests.

`uv run jupyter lab` — opens `notebooks/final.ipynb`, a guided walkthrough of the same steps.

References

[1] S. P. Bhat and D. S. Bernstein. A topological obstruction to continuous global stabilization of rotational motion and the unwinding phenomenon. Systems & Control Letters, 39(1):63–70, 2000.

[2] S. Boyd, L. El Ghaoui, E. Feron, and V. Balakrishnan. Linear Matrix Inequalities in System and Control Theory. SIAM, 1994.

[3] S. Diamond and S. Boyd. CVXPY: A Python-embedded modeling language for convex optimization. Journal of Machine Learning Research, 17(83):1–5, 2016.

[4] H. K. Khalil. Nonlinear Systems. Prentice Hall, 3rd edition, 2002.

[5] T. Lee. Exponential stability of an attitude tracking control system on SO(3) for large-angle rotational maneuvers. Systems & Control Letters, 61(1):231–237, 2012.

[6] T. Lee, M. Leok, and N. H. McClamroch. Geometric tracking control of a quadrotor UAV on SE(3). In 49th IEEE Conference on Decision and Control, pages 5420–5425, 2010.

[7] E. Todorov, T. Erez, and Y. Tassa. MuJoCo: A physics engine for model-based control. In IEEE/RSJ International Conference on Intelligent Robots and Systems, pages 5026–5033, 2012.